

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A) Date: 27-03-26
Time: 3 Hrs Max. Marks: 300

27-03-26_SR.IIT_STAR CO-SC(MODEL-A)_JEE MAIN_SPCL GTM-14_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

Rankers Academy JEE
THE NARAYANA GROUP

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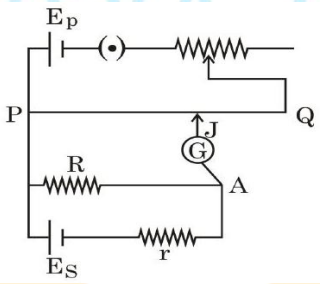
PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

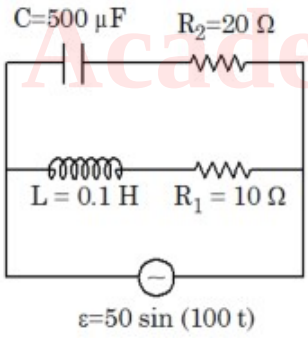
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. One of the circuits for the measurement of resistance by potentiometer is shown. The galvanometer is connected at point A and zero deflection is observed at length $PJ = 30$ cm . In second case the secondary cell is changed and the zero deflection is observed at length $PJ = 10$ cm. What is the resistance R (in ohm)? Take $E_S = 10\ V \& r = 1\Omega$ in 1st reading and $E_S = 5\ V \& r = 2\Omega$ in 2nd reading :-



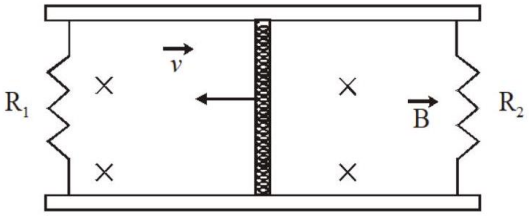
A) 1Ω B) 2Ω C) 3Ω D) 4Ω

2. In the circuit shown the average power developed in the resistor ' R_1 ' is



A) 31.25 W B) 62.50 W C) 125 W D) 250 W

3. A conducting bar of length L is free to slide on two parallel conducting rails as shown in the figure



Two resistors R_1 and R_2 are connected across the ends of the rails. There is a uniform magnetic field $\vec{B}$ pointing into the page. An external agent pulls the bar to the left at a constant speed v. The correct statement about the directions of induced currents I_1 and I_2 flowing through R_1 and R_2 respectively is :

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A) Both I_1 and I_2 are in anticlockwise direction

B) Both I_1 and I_2 are in clockwise direction

C) I_1 is in clockwise direction and I_2 is in anticlockwise direction

D) I_1 is in anticlockwise direction and I_2 is in clockwise direction

4. In a specially designed vernier calipers, 10 divisions of the vernier scale are equal to 4 divisions of the main scale. In the following figure, the zero error and a measurement are shown. In the zero error, the 5th and the 10th divisions of the vernier coincides with 1 mm and 3 mm mark on the main scale respectively and in the measurement, the 2nd and the 7th divisions of the vernier scale coincide with the 23 mm and 25 mm mark on the main scale respectively as shown in the figure. Report the length measured with due regards to the error.



A) (21.2 ± 0.2)mm B) (23.2 ± 0.2)mm C) (23.8 ± 0.4)mm D) (23.4 ± 0.6)mm

5. The electric field of a plane polarized electromagnetic wave in free space at time $t=0$ is given by an expression $\vec{E}(x,y)=10\hat{j}\cos\left[(6x+8z)\right]$. The magnetic field $\vec{B}(x,z,t)$ is given by : (c is the velocity of light)

A) $\frac{1}{c}\left(6\hat{k}+8\hat{i}\right)\cos\left[(6x-8z+10ct)\right]$ B) $\frac{1}{c}\left(6\hat{k}-8\hat{i}\right)\cos\left[(6x+8z-10ct)\right]$

C) $\frac{1}{c}\left(6\hat{k}+8\hat{i}\right)\cos\left[(6x+8z-10ct)\right]$ D) $\frac{1}{c}\left(6\hat{k}-8\hat{i}\right)\cos\left[(6x+8z+10ct)\right]$

6. The focal lengths of the objective and the eyepiece of a compound microscope are 2.0 cm and 3.0 cm respectively. The distance between the objective and the eye-piece is 15.0cm . The final image formed by the eye-piece is at infinity. The two lenses are thin. The distances in cm of the object and the image produced by the objective measured from the objective lens are respectively

A) 2.4 and 12.0 B) 2.4 and 15.0 C) 2.3 and 12.0 D) 2.3 and 3.0

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7. With respect to photoelectric experiment, match the entries of List-I with the entries of List-II.

LIST-I			LIST-II
P)	If ν (frequency) is increased keeping I (intensity) and ϕ (work function) constant	1)	Stopping potential increases
Q)	If I is increased keeping ν and ϕ constant	2)	Saturation photocurrent increases
R)	If the distance between anode and cathode increases.	3)	Maximum KE of the photoelectrons increase
S)	If ϕ is decreased keeping ν and I constant	4)	Stopping potential remains the same

A) $P \rightarrow 1,3; Q \rightarrow 2,4; R \rightarrow 2; S \rightarrow 2,4$ B) $P \rightarrow 1,3; Q \rightarrow 2,4; R \rightarrow 4; S \rightarrow 1,3$

C) $P \rightarrow 3; Q \rightarrow 1,2; R \rightarrow 2,4; S \rightarrow 1$ D) $P \rightarrow 2,3; Q \rightarrow 2; R \rightarrow 1,3; S \rightarrow 1,4$

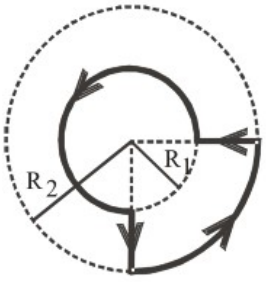
8. The wavelengths of K_α x-rays of two metals 'A' and ' B ' are $\frac{4}{1875R}$ and $\frac{1}{675R}$ respectively, where ' R ' is ryberg constant. The number of elements lying between ' A ' and ' B ' according to their atomic numbers is :

A) 3 B) 6 C) 5 D) 4

9. Binding Energy per nucleon of a fixed nucleus X^A is 6 MeV . It absorbs a neutron moving with $KE = 2MeV$, and converts into Y at ground state, emitting a photon of energy 1 MeV . The Binding Energy per nucleon of Y (in MeV) is

A) $\frac{(6A+1)}{(A+1)}$ B) $\frac{(6A-1)}{(A+1)}$ C) 7 D) $\frac{7}{6}$

10. What is the magnetic field at the common centre of the wire circuit (continuous line) as shown in the figure below?

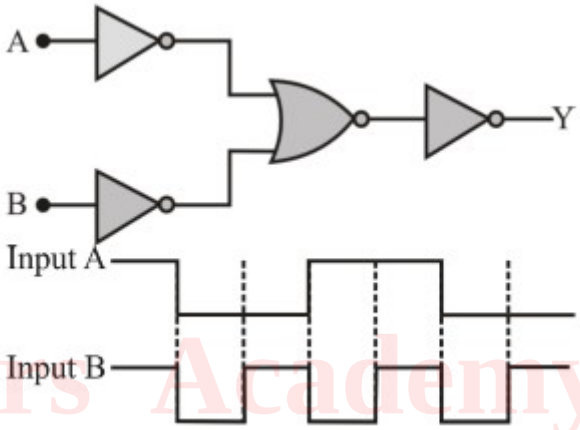


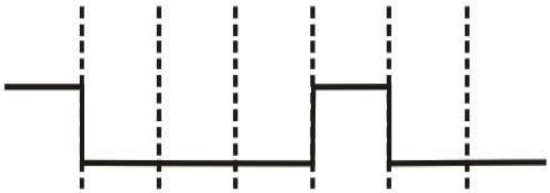
A) $\frac{\mu_0 I}{8}\left[\frac{1}{R_1}+\frac{1}{R_2}\right]$ B) $\frac{\mu_0 I}{8}\left[\frac{3}{R_1}+\frac{1}{R_2}\right]$ C) $\frac{\mu_0 I}{8}\left[\frac{1}{R_1}+\frac{3}{R_2}\right]$ D) $\frac{\mu_0 I}{8}\left[\frac{3}{R_1}-\frac{1}{R_2}\right]$

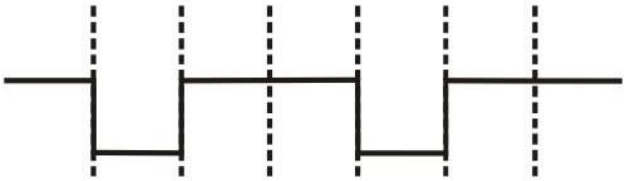
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11. A coil is placed in magnetic field such that plane of coil is perpendicular to the direction of magnetic field. The magnitude of magnetic flux through a coil can be changed:
A) By changing the magnitude of the magnetic field within the coil.
B) By changing the area of coil within the magnetic field.
C) By changing the angle between the direction of magnetic field and the plane of the coil.
D) By reversing the magnetic field direction abruptly without changing its magnitude.
Choose the most appropriate answer from the options given below:
A) A and B only
B) A, B and C only
C) A, B and D only
D) A and C only

12. The logic circuit shown has the input waveforms 'A' and 'B' as shown. Pick out the CORRECT output waveform :-



A)


B)


C)


D)


13. An unpolarized light passes through three polarising sheets whose polarising directions make an angle of $30^{\circ}, 60^{\circ}$ and 30° with y axis in same sense. What fraction of initial intensity is transmitted by the system?
A) $\frac{1}{2}$
B) $\frac{9}{32}$
C) $\frac{3}{32}$
D) $\frac{9}{64}$

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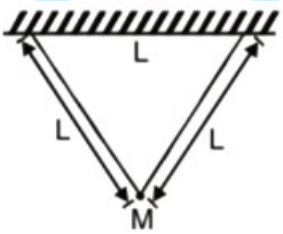
14. In YDSE, find the thickness of a glass slab ($\mu = 1.5$) which should be placed before the upper slit so that the central maximum now lies at a point where 5th bright fringe was lying earlier (before inserting the slab). Wavelength of light used is 5000Å

A) 5×10^{-6} m B) 3×10^{-6} m C) 10×10^{-6} m D) 5×10^{-5} m

15. If the mass, time and work are tanken as fundamental physical quantities then dimensional formula of length will be

A) $M^{1/2}\ T^{1}\ W^{-1/2}$ B) $M^{-1/2}\ T^{1}\ W^{1/2}$ C) $M^{-1}\ T^{2}\ W^{1/2}$ D) None of these

16. A man is swinging on a swing made of 2 ropes of equal length L and in direction perpendicular to the plane of paper. The time period of the small oscillations about the mean position is

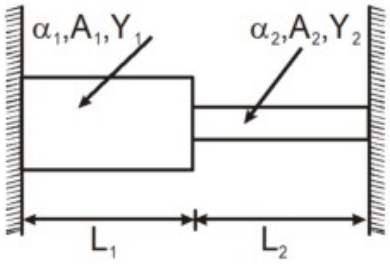


A) $2\pi\sqrt{\frac{L}{2g}}$ B) $2\pi\sqrt{\frac{\sqrt{3}L}{2g}}$ C) $2\pi\sqrt{\frac{L}{2\sqrt{3}g}}$ D) $\pi\sqrt{\frac{L}{\sqrt{g}}}$

17. Two rods are joined between fixed supports as shown in the figure. Condition for no change in the lengths of individual rods with the increase of temperature will be ($\alpha_1, \alpha_2 =$ linear expansion co-efficient

$A_1,\ A_2 =$ Area of rods

$Y_1, Y_2 =$ Young modulus)



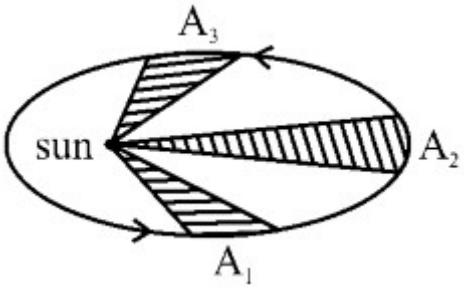
A) $\frac{A_1}{A_2}=\frac{\alpha_1 Y_1}{\alpha_2 Y_2}$ B) $\frac{A_1}{A_2}=\frac{L_1\alpha_1 Y_1}{L_2\alpha_2 Y_2}$ C) $\frac{A_1}{A_2}=\frac{L_2\alpha_2 Y_2}{L_1\alpha_1 Y_1}$ D) $\frac{A_1}{A_2}=\frac{\alpha_2 Y_2}{\alpha_1 Y_1}$

18. A uniform rod AB of mass m and length ℓ is at rest on a smooth horizontal surface. An impulse J is applied to the end B perpendicular to the rod in horizontal direction. Speed of the point A of the rod after giving impulse is :

A) $2\frac{J}{m}$ B) $\frac{J}{\sqrt{2}\,m}$ C) $\frac{J}{m}$ D) $\sqrt{2}\frac{J}{m}$

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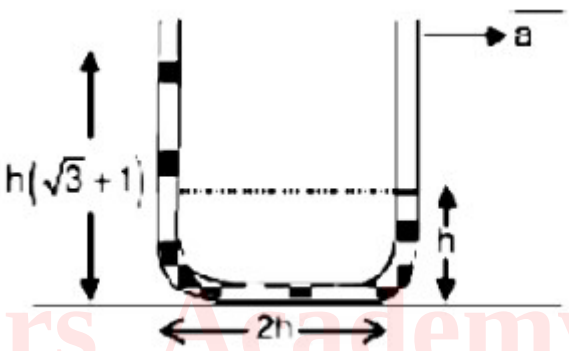
19. A planet moving around sun sweeps area A_1 in 2 days, A_2 in 3 days and A_3 in 6 days. Then the relation between A_1, A_2 and A_3 is :-



A) $3A_1 = 2A_2 = A_3$ B) $2A_1 = 3A_2 = 6A_3$

C) $3A_1 = 2A_2 = 6A_3$ D) $6A_1 = 3A_2 = 2A_3$

20. Liquid in U tube is shown. The acceleration of the tube a is :

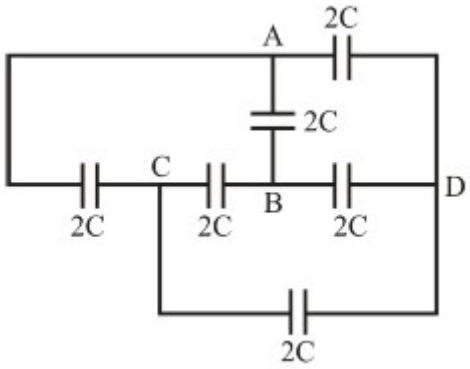


A) $\frac{g}{2}$ B) $\frac{\sqrt{3}\,g}{2}$ C) $\sqrt{3} \times g$ D) $\frac{\sqrt{3}}{2} \times g$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

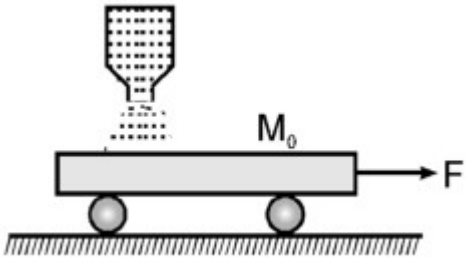
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. Equivalent capacitance between A and B is nC . The value of n is :



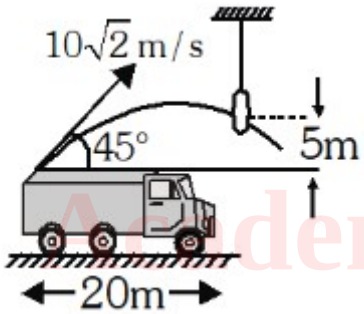
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22. In the figure shown a cart moves on a smooth horizontal surface with a constant velocity v acted upon by an external constant force of magnitude F. Sand falls on to the cart with negligible velocity at constant rate μkg/s and sticks to the cart. Find N if the power supplied by external agent applying the force F is μv^N.



23. In the figure shown, the trolley is moving with constant velocity (w×10)m/s. The boy is standing in the trolley throws a ball with velocity $10\sqrt{2}$ m/s at an angle 45° with horizontal with respect to trolley. Find W for which ball passes through the loop.

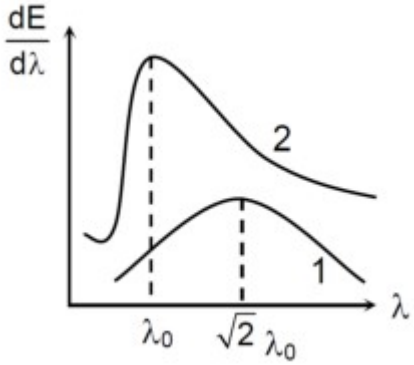
($g = 10\ m/s^2$)



24. An alluminium vessle of mass 0.5 kg contains 0.2 kg of water at $20^\circ C$. A block of iron of mass 0.2 kg at $100^\circ C$ is gently put it to water. If the equilibrium temperature of mixture is $T^\circ C$. Fill the value of $\frac{T}{5}$ in OMR. $S_{iron} = 4.70\ kg^{-1}k^{-1}$;

$S_{AC} = 910\ J\ kg^{-1}k^{-1}; S_{water} = 4200\ J\ kg^{-1}k^{-1}$.

25. Graph 1 and 2 represent the spectral emissive power of a body at temperature T_1 and T_1. The ratio of area under curve 2 and curve 1 is.



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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Which orbitals are filled in given order in 5th period of perodic table :-

A) $5s \rightarrow 4d \rightarrow 5p$ B) $5s \rightarrow 4d \rightarrow 5d$ C) $4s \rightarrow 5d \rightarrow 4p$ D) $5s \rightarrow 3d \rightarrow 4p$

27.

	Column I (Atomic number)		**Column II** (Block)
(P)	8	(1)	d-block
(Q)	12	(2)	p-block
(R)	60	(3)	s-block
(S)	26	(4)	f-block

	P	Q	R	S
(A)	1	3	2	4
(B)	2	3	4	1
(C)	3	4	1	2
(D)	4	1	3	2

A) A B) B C) C D) D

28. Sodium carbonate extract is a mixture of :

A) $[\text{Salt} + Na_2CO_3 + HCl]$ B) $[\text{Salt} + Na_2CO_3 + H_2O]$

C) $[\text{Salt} + CaCO_3 + HCl]$ D) $[\text{Salt} + Na_2CO_3 + HNO_3]$

29. In an alkaline solution, sodium nitroprusside gives a violet colour with :

A) SO_4^{2-} B) S^{2-} C) NO_2^- D) $S_2O_3^{2-}$

30. Maximum number of 2c-2e bonds which are formed by $sp^3 - sp^3$ overlapping in $Al_2(CH_3)_6$

A) 2 B) 4 C) 6 D) 8

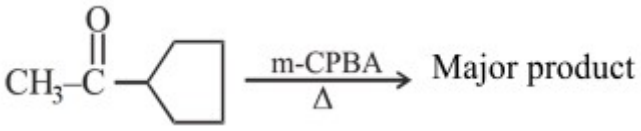
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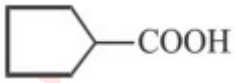
31. Which of the following statements are correct about HCN and HCP .
(I) HCN is linear with triple bond between carbon and nitrogen
(II) HCP does not exist because P does not form triple bond due to large size
(III) In HCP molecule two s-bond and two p bonds are present
(IV) Carbon atom is sp hybridised in HCN
A) Only I B) I and IV C) I, II, IV D) I, III and IV

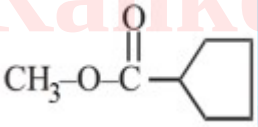
32. Find the difference in E.A.N. of the complexes $\left[Ti(\sigma-C_5H_5)_2(\pi-C_5H_5)_2\right]$ and $\left[Cr(C_6H_6)_2\right]$
A) 2 B) 4 C) zero D) 3

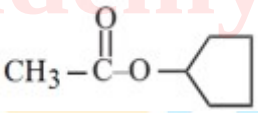
33. Which of the following obey Sidgwick EAN rule?
A) Wilkinson catalyst B) cis-platin
C) Brown ring complex D) Prussian blue

34. Which of the following represent major product for the given reaction

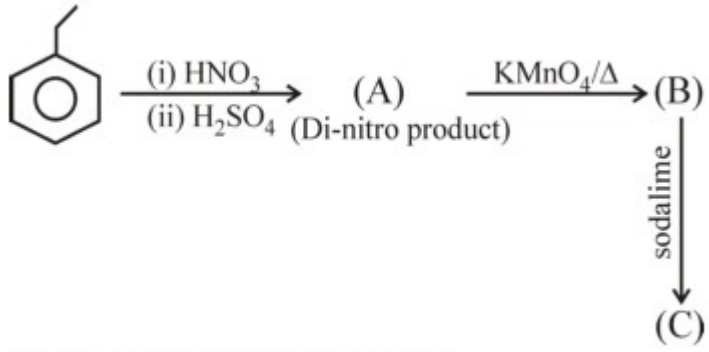


A) CH_3-COOH B) 

C) 

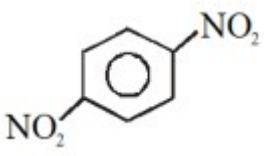
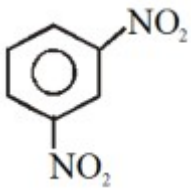
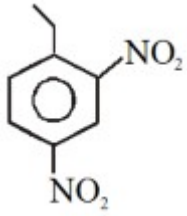
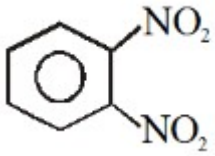
D) 

35. $CH_3-Br \xrightarrow[DMF]{KCN} X \xrightarrow[(ii)H_3O^+]{(i)\ PhMgCl} Y \xrightarrow{LiAlH_4} Z$ Z is :
A) $Ph-CH_2-OH$ B) $Ph-\underset{\underset{OH}{|}}{CH}-Ph$
C) $Ph-CH_2-CH_2-OH$ D) $Ph-\underset{\underset{OH}{|}}{CH}-CH_3$

36. 

Product (C) of this reaction is :-

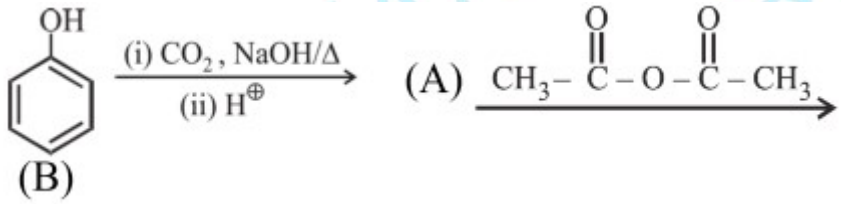
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A)  B)  C)  D) 

37. Which of the following aromatic compound is most reactive towards electrophilic addition reaction :

A)Toluene B) Benzaldehyde

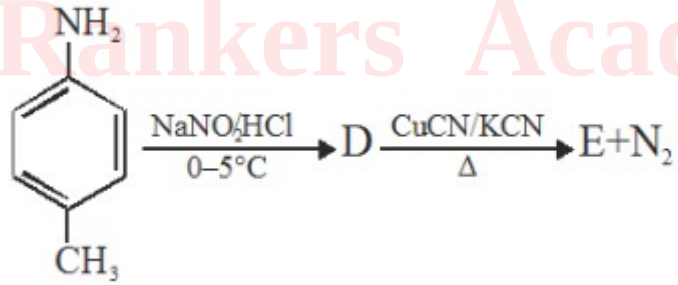
C) Methoxy benzene D) Benzene

38. 

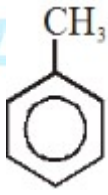
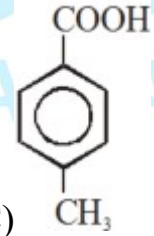
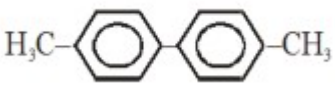
product B is :

A) Antibiotic B)Analgesic C) Antacid D) Insecticide

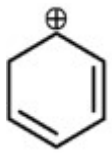
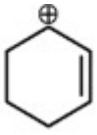
39. In the reaction



the product E is : —

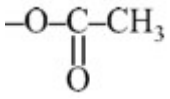
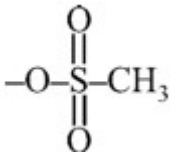
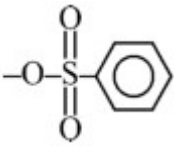
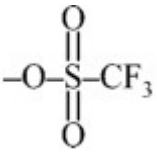
A)  B)  C)  D) 

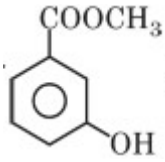
40. Which of the following carbocation is less stable than $CH_3-\overset{\oplus}{C}H-CH_3$:

A)  B)  C)  D) 

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41. Group having highest leaving tendency is :

A)  B)  C)  D) 

42. The IUPAC name of  is :

A) 5-Hydroxy benzene carboxylate B) 3-Hydroxy benzene carboxylate

C) 3-Methoxy carbonyl benzene-1-ol D) Methyl 3-hydroxy benzoate

43. An equilibrium mixture of $NO_2(g)$ and $N_2O_4(g)$ is present in a closed container at 300 K with pressures 0.4 atm and 0.2 atm respectively. On doubling the volume of container, the pressure of $NO_2(g)$ at new equilibrium at 300 K will be

A) 0.19 atm B) 0.35 atm C) 0.2 atm D) 0.24 atm

44. 0.25 mole of solid " A " is taken in a 1 litre evacuated flask at $27^\circ C$. Following two equilibrium exist simultaneously.

$$A(s) \rightleftharpoons B(g) + C(g) \ldots\ldots (i) K_{C_1}$$

$$2B(g) \rightleftharpoons D(g) + 2E(g) \ldots\ldots (ii) K_{C_2}$$

Let 0.2 moles of solid decomposes upto equilibrium & 0.1 mole of E was found at equilibrium.

[Given : $R = 0.08$ L atm K^{-1} mol^{-1}]

Calculate partial pressure of B at equilibrium.

A) 1.2 atm B) 0.8 atm C) 2.4 atm D) None of these

45. Find which pair do not belong to isotones.

A) ${}_6C^{13}$ & ${}_7N^{14}$ B) ${}_{15}P^{31}$ & ${}_{16}S^{32}$ C) ${}_{14}Si^{30}$ & ${}_{15}P^{31}$ D) ${}_5B^{11}$ & ${}_6C^{13}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:

Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. Calculate molar entropy change

(in cal deg $^{-1}$ mole $^{-1}$) in following state change

$$H_2O(s, -10^\circ C, 1 \text{ atm}) \rightarrow H_2O(\ell, 10^\circ C, 1 \text{ atm})$$

C_p for $H_2O(s) = 10 \text{ cal deg}^{-1} \text{ mole}^{-1}$

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C_p for $H_2O(\ell) = 18 cal deg^{-1}\ mole^{-1}$

$\left[\text{Take}: \ln\left(\frac{273}{263}\right) = 0.037, \ln\left(\frac{283}{273}\right) = 0.035\right]$

Latent heat of fusion of $H_2O(s) = 1.092 kcal/mole 0^\circ C$.

47. Copper metal reacts with nitric acid. Assume that the reaction

$3Cu(s) + 8HNO_3(aq) \rightarrow 3Cu(NO_3)_2(aq) + 2NO(g) + 4H_2O(l)$

If $18.75\ g Cu(NO_3)_2$ is eventually obtained, then how many grams of nitric oxide, NO, would have also formed according to the preceeding equation?

48. Calculate e.m.f of cell

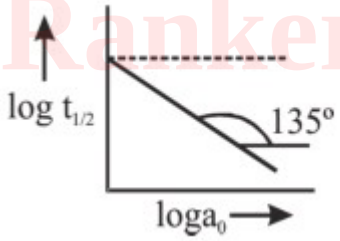
$Ag|AgNO_3(0.1M)||Ag(NH_3)_2^+(0.01M),\ NH_3(1M)|Ag$

Given : - $K_f\left(\left[Ag(NH_3)_2^+\right]\right) = 10^8$

$\frac{2.303RT}{F} = 0.06$

49. A graph between $\log t_{1/2}$ vs $\log a_0$, where a_0 is initial concentration of reactant A in the reaction A → product

is shown



The rate law is $-\frac{dA}{dt} = k[A]^n$ Find the value of ' n '.

50. For the inversion of cane sugar $(C_{12}H_{22}O_{11})$ obeying I order following data were obtained

Time (sec.)	0	10	∞
Angle of rotation of solution (degree)	+20	- 2.5	- 10

What will be rate constant in $sec^{-1}(\ell n 2 = 0.7)$

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Four of eight vertices of a regular octagon are chosen at random. Then the probability that the quadrilateral formed is square is :-

A) $\frac{1}{35}$ B) $\frac{2}{35}$ C) $\frac{3}{35}$ D) $\frac{4}{35}$

52. A straight line passes through a fixed point (h, k). The locus of the foot of perpendicular on it drawn from the origin is

A) $x^2+y^2-hx-ky=0$ B) $x^2+y^2+hx+ky=0$

C) $3x^2+3y^2+hx-ky=0$ D) None of these

53. If the domain of $f(x)$ is $[-1,2]$ then the domain of the function $f([x]-x^2+4)$ is (where [.] denotes greatest integer function) :-

A) $[-1,\sqrt{7}]$ B) $[-\sqrt{3},-1]\cup[\sqrt{3},\sqrt{7}]$

C) $[-\sqrt{3},\sqrt{7}]-(-1,2)$ D) $[\sqrt{13},\sqrt{15}]$

54. If $\ell=10\sec^2 x+9\cos^2 x, m=3\tan^2 y+12\cot^2 y$ and $n=7\cos 2z-24\sin 2z+25$, then number of common possible integral values of ℓ, m and n are

A) 30 B) 31 C) 32 D) 33

55. If $\hat{a},\hat{b}$ and $\hat{c}$ are unit vectors, then $|\hat{a}-\hat{b}|^2+|\hat{b}-\hat{c}|^2+|\hat{c}-\hat{a}|^2$ does not exceed :-

A) 4 B) 9 C) 8 D) 6

56. If the matrices $A, B, (A+B)$ are non-singular, then $\left[A(A+B)^{-1}B\right]^{-1}$ is equal to -

(A) $A+B$ (B) $A^{-1}+B^{-1}$ (C) $(A+B)^{-1}$ (D) $B^{-1}A+BA^{-1}$

57. The value of $\cot\left(\sum_{n=1}^{23}\cot^{-1}\left(1+\sum_{k=1}^{n}2k\right)\right)$ is

A) $\frac{23}{25}$ B) $\frac{25}{23}$ C) $\frac{23}{24}$ D) $\frac{24}{23}$

58. Let $S_n=1\cdot(n-1)+2\cdot(n-2)+3\cdot(n-3)+....+(n-1)\cdot 1$. The sum $\sum_{n=4}^{\infty}\left(\frac{2S_n}{n!}-\frac{1}{(n-2)!}\right)$ is equal to

A) $\frac{e-1}{3}$ B) $\frac{e-2}{6}$ C) $\frac{e}{3}$ D) $\frac{e}{6}$

59. Three circles of radii $a,b,c(a<b<c)$ touch each other externally. If they have x -axis as a common tangent, then :

A) $\frac{1}{\sqrt{a}}=\frac{1}{\sqrt{b}}+\frac{1}{\sqrt{c}}$ B) a,b,c are in A.P.

C) $\sqrt{a},\sqrt{b},\sqrt{c}$ are in A.P. D) $\frac{1}{\sqrt{b}}=\frac{1}{\sqrt{a}}+\frac{1}{\sqrt{c}}$

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60. Let $\beta=\lim_{x\to0}\frac{\alpha x-\left(e^{3x}-1\right)}{\alpha x\left(e^{3x}-1\right)}$ for some $\alpha\in\mathbb{R}$. Then the value of $\alpha+\beta$ is :

A) $\frac{14}{5}$ B) $\frac{3}{2}$ C) $\frac{5}{2}$ D) $\frac{7}{2}$

61. The remainder when $(2021)^{2022}+(2022)^{2021}$ is divided by 7 is

A) 0 B) 1 C) 2 D) 6

62. If $f(x)=\begin{cases}\int_0^x(5+|1-t|)\,dt, & x>2\\ 0 & x\leqslant2\end{cases}$, then

A) $f(x)$ is not continuous at $x=2$

B) $f(x)$ is everywhere differentiable

C) $f(x)$ is continuous but not differentiable at $x=2$

D) $f(x)$ is not differentiable at $x=1$

63. The locus of the mid-point of the line segment joining the focus of the parabola $y^2=4ax$ to a moving point of the parabola, is another parabola whose directrix is :

A) $x=-\frac{a}{2}$ B) $x=\frac{a}{2}$ C) $x=0$ D) $x=a$

64. Let $f:\mathbb{R}\to\mathbb{R}$ be a function defined by $f(x)=(x-3)^{n_1}(x-5)^{n_2}, n_1,n_2\in N$. Then which of the following is NOT true ?

A) For $n_1=3,n_2=4$, there exists ' a ' $\in(3,5)$ where fattains local maxima at $x=a$

B) For $n_1=4,n_2=3$, there exists ' a ' $\in(3,5)$ where f attains local minima at $x=a$

C) For $n_1=3,n_2=5$, there exists ' a ' $\in(3,5)$ where f attains local maxima at $x=a$

D) For $n_1=4,n_2=6$, there exists ' a ' $\in(3,5)$ where f attains local maxima at $x=a$

65. If $x^2+9y^2-4x+3=0, x,y\in\mathbb{R}$, then x and y respectively lie in the intervals:

A) $\left[-\frac{1}{3},\frac{1}{3}\right]$ and $\left[-\frac{1}{3},\frac{1}{3}\right]$ B) $\left[-\frac{1}{3},\frac{1}{3}\right]$ and $[1,3]$

C) $[1,3]$ and $[1,3]$ D) $[1,3]$ and $\left[-\frac{1}{3},\frac{1}{3}\right]$

66. $\int\frac{x^3}{\left(2x^2+1\right)^3}dx$ is equal to- (where ' C' is integration constant)

A) $\frac{1}{4}\left(2+\frac{1}{x^2}\right)^{-2}+C$ B) $-\frac{1}{4}\left(2+\frac{1}{x^2}\right)^{-2}+C$

C) $\frac{1}{2}\left(2+\frac{1}{x^2}\right)^{-2}+C$ D) $\frac{1}{4}\left(2+\frac{1}{x^2}\right)^{2}+C$

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67. If the lines $\frac{x-1}{2}=\frac{y-2}{0}=\frac{z+1}{k}$ and $\frac{x+3}{1}=\frac{y-1}{k}=\frac{z-k}{-2}$ are coplaner, then

A) No real value of k exist
B) Exactly one real value of k exists
C) Exactly two real value of k exists
D) Exactly three real value of k exists

68. If a and b are two numbers such that $a^2+b^2=7$ and $a^3+b^3=10$, then

A) greatest value of $|a+b|$ is 4
B) greatest value of $|a+b|$ is 10
C) smallest value of $|a+b|$ is 1
D) smallest value of $|a+b|$ is 2

69. The area of region of the xy plane defined by the inequality $|x|+|y|+|x+y|\le 1$ is

A) $1/2$ B) $3/4$ C) 1 D) 2

70. The mean and variance of a set of 15 numbers are 12 and 14 respectively. The mean and variance of another set of 15 numbers are 14 and σ^2 respectively. If the variance of all the 30 numbers in the two sets is 13 , then σ^2

A) 9 B) 12 C) 11 D) 10

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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71. The number of odd positive integral solutions of the equation $x+y+z=53$ is N , then $\frac{N}{100}$ is :

72. If $m=\int_{-1}^{2} x\sin\left(x(1-x)\right)dx$ and $n=\int_{-1}^{2}\sin\left(x(1-x)\right)dx$, then $\frac{m}{n}$ is equal to :-

73. If $\sin^6\theta$ & $\cos^6\theta$ are the roots of equation $10x^2-5x+\lambda_1=0$ and $\sin^4\theta$ & $\cos^4\theta$ are roots of equation $x^2-\lambda_2 x+\lambda_3=0$, then $36\lambda_1+\lambda_2+24\lambda_3$ is equal to

74. If $z^2+z+1=0$, where z is complex number then the value of

$$\left(z+\frac{1}{z}\right)^3+\left(z^2+\frac{1}{z^2}\right)^3+\left(z^3+\frac{1}{z^3}\right)^3+\ldots+\left(z^7+\frac{1}{z^7}\right)^3$$

75. If p,q,r are in A.P., then the value of determinant $\begin{vmatrix} k+2 & k+3 & k+4 \\ k+3 & k+4 & k+5 \\ k+2p & k+2q & k+2r \end{vmatrix}$ is equal to :

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